From the Figure 7.39, $\tan\beta = \frac{h}{f_o} = \frac{h'}{L}$, then

$$\frac{h'}{h} = \frac{L}{f_o} \quad (7.74)$$

$$m_o = \frac{L}{f_o} \quad (7.75)$$

Here, the distance $L$ is measured between the focal point of the eyepiece to the focal point of the objective. This is called the tube length of the microscope as $f_o$ and $f_e$ are comparatively smaller than $L$.

If the final image is formed at the near point, the magnification $m_e$ of the eyepiece is,

$$m_e = 1 + \frac{D}{f_e} \quad (7.76)$$

The total magnification $m$ for near point focusing is,

$$m = m_o m_e = \left(\frac{L}{f_o}\right)\left(1 + \frac{D}{f_e}\right) \quad (7.77)$$

If the final image is formed at infinity (normal focusing), the magnification $m_e$ of the eyepiece is,

$$m_e = \frac{D}{f_e} \quad (7.78)$$

The total magnification $m$ for normal focusing is,

$$m = m_o m_e = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right) \quad (7.79)$$

## EXAMPLE 7.20

A microscope has an objective and eyepiece of focal lengths 5 cm and 50 cm respectively with tube length 30 cm. Find the magnification of the microscope in the (a) near point and (b) normal focusing.

### Solution

$f_0 = 5\text{cm} = 5\times10^{-2}\text{m}$; $f_e = 50\text{cm} = 50\times10^{-2}\text{m}$;

$L = 30\text{cm} = 30\times10^{-2}\text{m}$; $D = 25\text{cm} = 25\times10^{-2}\text{m}$

(a) The total magnification $m$ in near point focusing is, $m = m_o m_e = \left(\frac{L}{f_o}\right)\left(1 + \frac{D}{f_e}\right)$

Substituting,

$$m = m_o m_e = \left(\frac{30\times10^{-2}}{5\times10^{-2}}\right)\left(1 + \frac{25\times10^{-2}}{50\times10^{-2}}\right)$$

$$= (6)(1.5) = 9$$

(b) The total magnification $m$ in normal focusing is, $m = m_o m_e = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$

Substituting,

$$m = m_o m_e = \left(\frac{30\times10^{-2}}{5\times10^{-2}}\right)\left(\frac{25\times10^{-2}}{50\times10^{-2}}\right)$$

$$= (6)(0.5) = 3$$

## 7.6.3 Astronomical telescope

An astronomical telescope is used to get the magnification of distant astronomical objects like stars, planets, moon etc. The image formed by astronomical telescope will be inverted. It has an objective of long focal length and a much larger aperture than the eyepiece as shown in Figure 7.40. Light from a distant object enters the objective and a real image is formed in the tube at its focal point. The eyepiece magnifies this image producing a final inverted image.

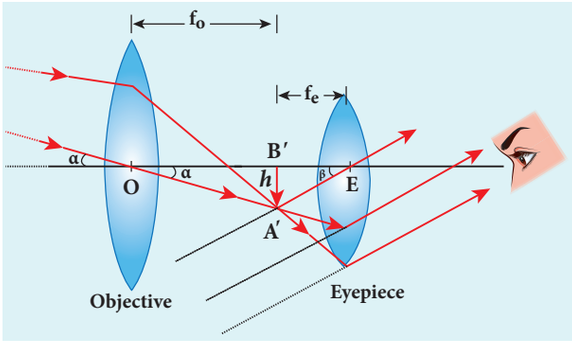



**Figure 7.40** Astronomical telescope



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